김동영 (金炯瑩) 이력서

과학과 공학을 위한 인공지능 및 빅 데이터 연구

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경력

2017 – 현재	한국 **기초과학연구원 (Institute for Basic Science)**, 첨단연성물질 연구단 연구요원 (Research Fellow), 지도교수: 조윤경 (UNIST).
2012 – 2016	미국 **Texas A&M University** 연구원 (Research Assistant) 지도교수: Raimund Ober (Texas A&M University).
2010 – 2012	미국 **The University of Southwestern medical center** 연구원 (Research Assistant)
2010 – 2012	미국 **The University of Texas at Dallas** 연구원 (Research Assistant)
2009 – 2010	**임베디드소프트웨어 연구센터** 연구원 (Research Assistant)
2010 – 2010	호주 **EOS Australia** 인턴쉽

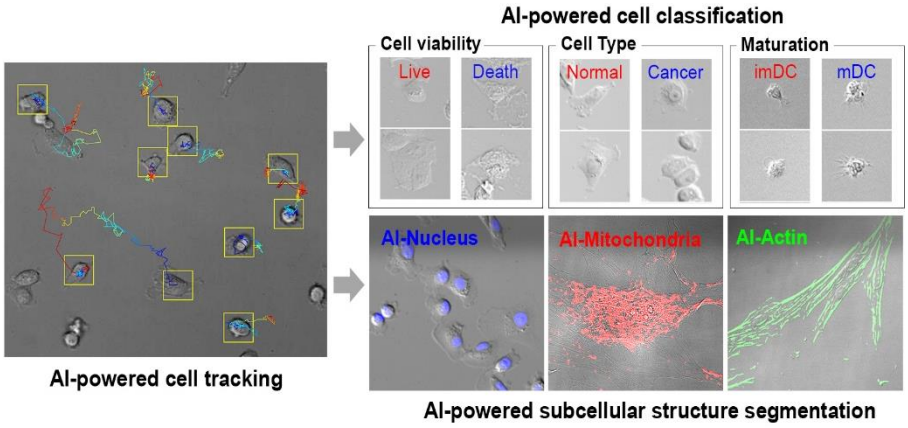
학력

2012 – 2016	미국 **Texas A&M University 의공학 박사 (Ph.D.)** College station, Texas, USA
2010 – 2012	미국 **The University of Texas at Dallas 전자공학 학사 (B.S.)** Richardson, Texas, USA
2006 – 2012	**경북대학교 전자 컴퓨터 공학 학사** 대구, 대한민국

프로젝트

2017 – 2019

인공지능 현미경 (AI-powered transmitted light microscopy for functional analysis of live cells).



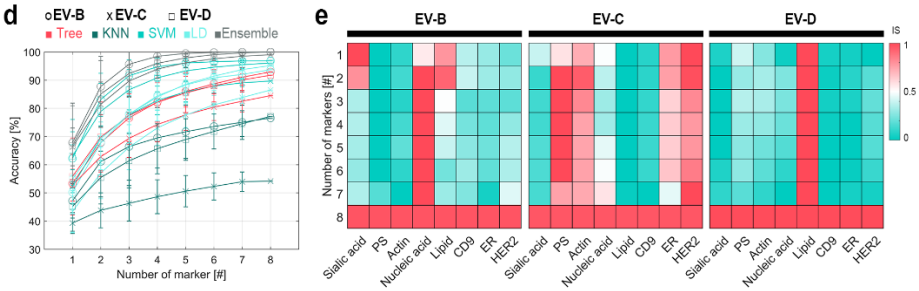
- Unsupervised machine learning 과 supervised semantic segmentation 을 이용한 transmitted light microscopy image 를 fluorescence microscopy image 로 바꾸는 기술.
- Region proposal convolutional neural network 및 convolutional neural network 를 이용한 transmitted light microscopy image 내의 세포 상태 실시간 모니터링.
- Time-lapse 영상 내의 세포를 complementary learner 를 이용한 tracking 및 모니터링.

*2 건의 국내 특허 출원 중.

*Manuscript under review (Sci. Rep. 2019).

2017 – 2019

혈액내 나노소포체 분석 및 기계학습을 통한 암 진단 (Find cancer-related particles from blood using machine learning)

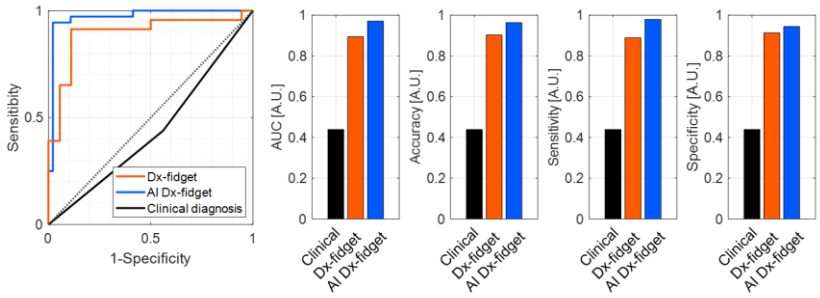


- 혈액내 나노소포체에 대해 Machine learning classification 을 통한 암진단.

*Manuscript under review (Anal. Chem. 2019).

2017 – 2019

인공신경망을 사용한 소변내 박테리아 감염 진단 (Point-of-care-testing urinary tract infection diagnosis using deep neural network)



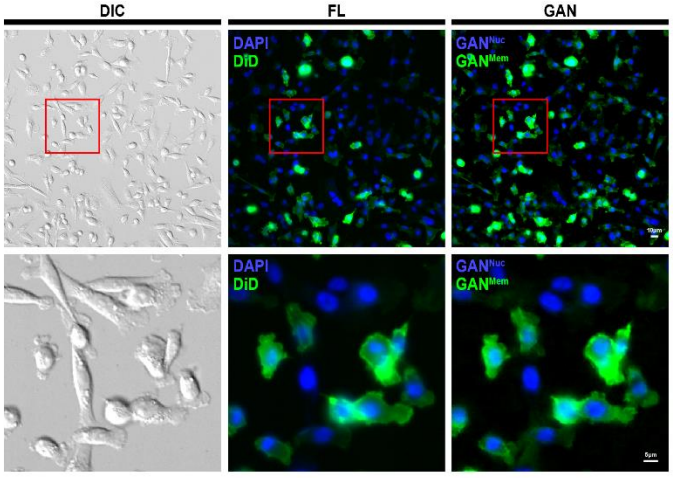
- Sample kit 의 휴대폰 사진으로부터 convolutional neural network 를 이용한 정상 및 박테리아 감염 환자 진단.

*1 건의 국내 특허 출원 중.

*Manuscript under review (Nature BME. 2019).

2019 – 현재

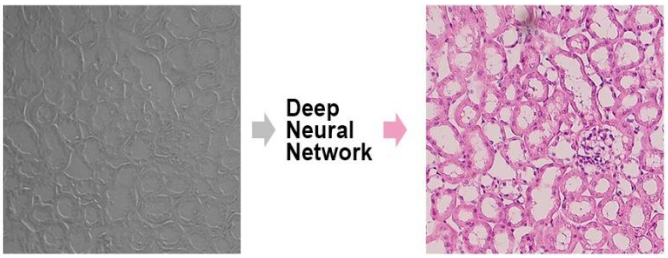
인공신경망을 이용한 형광 표식변환 (In-silico fluorescence staining using generative adversarial network).



*Manuscript under preparation (2019).

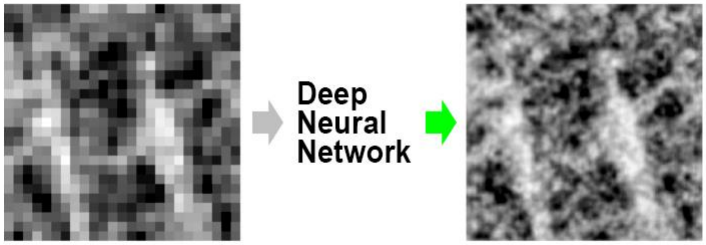
2019 – 현재

인공신경망을 이용한 조직학 이미지 생성 (In-silico histology staining using generative adversarial network).



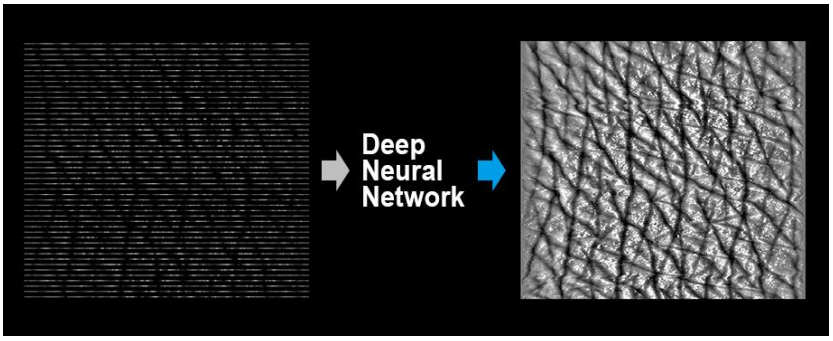
*Manuscript under preparation (2019).

2019 – 현재 **인공신경망을 통한 고해상도 의료영상 복원 (Generative adversarial network for high definition medical image).**



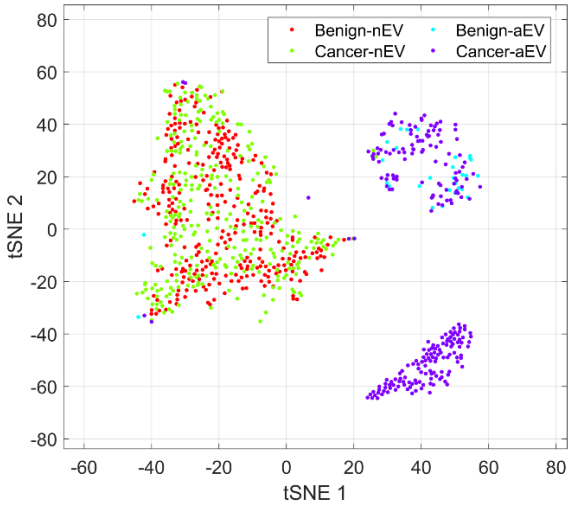
*Manuscript under preparation (2019).

2019 – 현재 **인공신경망을 통한 초고속 의료영상 촬영 (Generative adversarial network for rapid medical image acquisition).**



*Manuscript under preparation (2019).

2019 – 현재 **인공신경망을 통한 희귀 암 소포체 탐색 (Searching extreme rare cancer extracellular vesicle in blood plasma using generative adversarial network).**



*Manuscript under preparation (2019).

특허

김동영, 민유홍, 조윤경, 인공신경망을 이용한 특수 현미경 영상 생성 방법 및 영상 처리 장치 (1). 특허출원 10-2019-0001740 호, 한국특허출원, 2019.

김동영, 민유홍, 조윤경, 인공신경망을 이용한 세포 영상 분석 방법 및 세포 영상 처리 장치 (2). 특허출원 10-2019-0001741 호, 한국특허출원, 2019.

김동영, 미카엘아이작, 기동엽, 조윤경, 원심력 기반 무전원 입자 농축장치 및 입자 농축방법. 특허출원 제 10-2018-113085 호, 한국특허출원, 2018.

Kim D., Ober R.J, **Advanced multi-dimensional microscopy system for single particle & structure imaging.** U.S. Patent pending, 2017.

수상

2018	The 22nd International Conference On **Miniaturized Systems For Chemistry And Life Sciences**, Shark Tank Competition 1 위
2016	스위스 **SMLMS Challenge (EPFL)** 3D single molecule localization microscopy 부문 1 위
2014 – 2016	미국 **Texas A&M University** 연구 장학금
2012 – 2014	미국 **The University of Texas at Dallas** 연구 장학금
2011 – 2011	미국 **The University of Texas at Dallas** 학사 연구 장학금
2011 – 2011	미국 **The University of Texas at Dallas** Senior Design Project 2 위
2010 – 2012	미국 **The University of Texas at Dallas** 학사 장학금

언어

한국어 모국어, **영어** 유창함

기술

인공지능	Supervised/unsupervised machine learning, Deep neural network design, generative adversarial network design.
프로그래밍	MATLAB, Python, C, C++, JAVA.
데이터 처리	Big data analysis, Optimization based data processing.
시스템 설계	NI LabWindows/CVI, NI LabView.
광학설계	Zemax based optical system design, Multimodal optical system configuration.

현미경	형광 현미경 설계 및 제작 (Epifluorescence, TIRF microscopy, Confocal microscopy) 단일 분자 현미경/초고해상도 형광 현미경 설계 및 제작 (Single molecule microscopy, super-resolution microscopy) 전자현미경 사용 (Transmitted/scanning electron microscopy)

강의

BME21101 Introduction to biomedical engineering (**Introduction to artificial intelligence for biomedical engineering / Introduction to deep learning**), 2019, Ulsan National Institute of Science and Technology (UNIST).

논문

Kim. D., Min Y., Cho Y. **AI-powered transmitted light microscopy for functional analysis of live cells.** Scientific report 2019 (under review).

Kim. D., Woo H., Lee C,. Min Y., Cho Y. **Unveiling tumor specific extracellular vesicles by size fractionation and single vesicle analysis.** Analytical chemistry 2019 (under review).

Kim. D. Michael I**,** Oleksandra G., Kuimar S., Clara, J., Ki D., Cho Y. **Dx-Fidget Spinner: A Pocket-Sized Low-Cost Hand-Powered Medical Diagnostic Tool.** Nature Biomedical Engineering 2019 (under review).

Sunkara, V., Kim C., Park J., Woo H., **Kim D.**, Ha H., Kim M., Son Y., Kim J., Cho Y., **Fully automated, label-free isolation of extracellular vesicles from whole blood for cancer diagnosis and monitoring.** Theranostics, 2019.

Michael I., Kumar S., Oh J,, **Kim D**, Kim J., Cho Y. **Surface Engineered Paper Hanging Drop Chip for 3D Spheroid Culture and Analysis.** ACS applied materials & interfaces, 2018.

Li, R., Chiguru, S., Li, L., **Kim, D.**, Velmurugan, R., Kim, D., Tian, H., Schroit, A., Mason, R., Ober, R. J. and Ward, E. S. **Targeting phosphatidylserine with calcium-dependent protein-drug conjugates for the treatment of cancer.** Molecular Cancer Therapeutics, 2018.

Devanaboyina, S. C., Khare, P., Challa, **Kim, D.,** Ober, R. J., and Ward, E. S. **Engineered clearing agents for the selective depletion of antigen-specific antibodies.** Nature Communications, 2017.

Kim, D., Chao, J., Velmurugan, R., You, S., Ward, E. S., and Ober, R. J. **Remote focusing multifocal plane microscopy for the imaging of 3D single molecule dynamics with cellular context.** Proceedings of the SPIE, Three-Dimensional and Multidimensional Microscopy: Image Acquisition and Processing XXIV, 10070: 2017.

Vahid, M. R., Chao, J., **Kim, D.,** Ward, E. S., and Ober, R. J. **State space approach to single molecule localization in fluorescence microscopy.** Biomedical Optics Express, 2017

Poovasery J.S., Kang J.C., **Kim D**, Ober R.J, Ward E.S. **Antibody targeting of HER2/HER3 signaling overcomes heregulin-induced resistance to PI3K inhibition in prostate cancer.** International Journal of Cancer 2017

Kim D, You S, Ward E.S., Ober R.J, **Imaging of three-dimensional single molecule dynamics with cellular context: Antibody trafficking and interaction with cell membrane and sorting endosomes.** ASCB, San Fransisco, CA. 2016.

Cohen E. A. K.,. **Kim D**., Ober R.J, **Cramer-Rao Lower Bound for Point Based Image Registration with Heteroscedastic Error Model for Application in Single Molecule Microscopy.** IEEE Transactions on Medical Imaging 2015

Ram S, **Kim D**, Ober RJ, Ward ES. **The level of HER2 expression is a predictor of antibody-HER2 trafficking behavior in cancer cells**. mAbs 2014

Devanaboyina SC, Lynch SM, Ober RJ, Ram S, **Kim D**, Puig-Canto A, Breen S, Kasturirangan S, Fowler S, Peng L, et al. **The effect of pH dependence of antibody-antigen interactions on subcellular trafficking dynamics**. mAbs 2013

Ram S, **Kim D**, Ober RJ, Ward ES. **Microscopy in Polarized Epithelia Reveals a Novel Cellular Process of Intercellular Transfer.** Biophysical Journal 2013

Ram S, **Kim D**, Ober RJ, Ward ES. **3D Single Molecule Tracking with Multifocal Plane Microscopy Reveals Rapid Intercellular Transferrin Transport at Epithelial Cell Barriers**. Biophysical Journal 2012

Kim D, Ram S, Ober RJ, Ward ES. **3D single molecule tracking of rapid intracellular trafficking imaged by multifocal plane microscopy** Microscopy and Microanalysis 2012.